

Adithya Kalyan

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EDUCATION

University of Maryland - College Park

Graduating May 2027

B.S. | Computer Science and Mathematics (conc. AI/ML), Minor in Computational Finance

GPA 3.83 (Dean's List)

- Coursework: AI/ML, Advanced Data Structures, Algorithms, Data Science, Computational Methods, Probability Theory, Real Analysis, System Architecture, Computer & Network Security, Reinforcement Learning, Financial Markets & Datasets

EXPERIENCE

Software Engineer

September 2025 – April 2026

App Development Club - Robert H. Smith School of Business

College Park, MD

- Spearheaded student loan payment optimization and financial risk visualization through Monte-Carlo simulation for **iOS** app
- Implemented secure **Flask** user login and **API handling**; Engage users with games and course recommendation algorithm.

Quantitative Research Intern

August 2025 – March 2026

Shepherd Ventures

San Diego, CA

- Constructed beta forecasting strategies, leveraging **10+ years of daily time-series data** across **7 Black Swan** events.
- Engineered **7 risk matrices** for over 500 S&P stocks, enabling granular analysis of Swan events and risk fluctuation
- Applied supervised ML Hidden Markov Model trained on **21,000+ beta values** to mitigate risk during Black Swan events.
- Optimized unsupervised ML algorithm (**K-means Clustering**) with Davies-Bouldin Index; identified 8 stocks with lowest expected volatility and weighted accordingly, achieving **31.07%** protection rate compared against index performance.

Software Engineer Intern

May 2025 – August 2025

Boston Scientific

Minneapolis, MN

- Reduced engineer effort by **85%** per test case by building Azure OpenAI-powered **RAG** model to automate medical testing.
- Converted XML test protocols into **compilable** Python automation scripts; integrated into workflow for **50+** engineers.
- Automated XML-Python script parsing and mapping for database upload, handling **over 5,000 files weekly** to Azure
- Improved data-processing by **98.75%** (40 minutes to 30 seconds) while increasing script first-pass **accuracy by 30%**

Software Engineer Intern

May 2024 – October 2024

KeeperAI

New York, NY

- Implemented business value alignment scoring model enabling managers to proactively engage with their employees and driving a **15%** increase in quarterly employee engagement scores; feature achieved 90% adoption within first quarter.
- Saved developers **10 hours/week** by debugging recurring issues in recommendation engine and improving overall uptime.

PROJECTS

Flappy Bird - Zero Ping | [Repo](#) | C++, Threads, System Architecture

October - November 2025

- Developed low-latency physics engine using a **ThreadPool** to offload physics calculations and maintain stable **60 FPS**.
- Engineered Producer-Consumer **C++ concurrent pipeline** with **SafeQueue** to eliminate I/O blocking and in-game lag

Image Generation Diffusion Model | [Repo](#) | PyTorch, DDPM, UNet Architecture

June - July 2025

- Built Denoising Diffusion Probabilistic Model using **PyTorch** via **UNet backbone**; trained on **CIFAR10** dataset.
- Incorporated custom noise scheduling, forward/reverse sampling, image generation, and used MSE loss function.
- Engineered ML pipeline to enable high-quality image generation from Gaussian noise using a **cosine beta schedule**.

ML Trading Bot | [Repo](#) | Sentiment Analysis, Financial Markets, API handling

December 2024

- Developed **unsupervised learning** pipeline for financial **sentiment analysis**, tokenizing recent news headlines and using **PyTorch** to dynamically inform and drive the trading bot's core logic; used open source FinBERT NLP model.
- Engineered backtesting framework to validate trading strategies against S&P 500 by using Yahoo Finance historical data.
- Deployed trading strategies through **Alpaca API**, generating **22.8% annual returns** in real-time market conditions.

MiniOcaml Compiler | [Repo](#) | Ocaml, Dune, Regex

October – November 2024

- Designed and implemented a full compiler for MiniOCaml, encompassing both Lexer with optimal tokenization pipeline and robust CFG parser to create **AST** representation of MiniOcaml; handling variable binding, function definitions, etc.
- Developed core algorithms for formal language theory: NFA acceptance, **NFA-to-DFA** and regex-to-NFA translation.
- Created compile-time MiniOcaml evaluator with **constant folding** and propagation for enhanced execution efficiency.
- Eliminated all potential type errors with pre-runtime comprehensive **static type checker**; promoted compiler stability.

SKILLS

Languages: Python, Java, C/ C++, JavaScript, Rust, SQL, OCaml, MATLAB, Assembly, XML, HTML/CSS, Swift

Frameworks & Libraries: Pandas, PyTorch, Scikit-Learn, NumPy, Django, LangChain, React, Node.js, Flask, OpenCV

Developer Tools & Concepts: Azure, Git, Jira, CI/CD, AWS, Unix/Linux, REST APIs, Database Systems, Fusion 360

ECs/Interests: Startup Shell, App Dev Club, Varsity Golf Captain, Computer Science Tutor, Eagle Scouts, Photography